

CSMFAB00000000000019 — Aegis Home Stealth-Lift: Non-Inductive Industrial Forklift

Version 2.0 — Revised & Expanded | June 2026

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Δ Change Log — V1.0 □ V2.0

Rev	Date	Change Description
1.0	2024	Initial release — BFRP composite chassis, ZrO ₂ wheel hubs, piezoelectric/pneumatic actuators, YInMn Blue, ceramic battery enclosure
2.0	June 2026	BMW i3 CFRP/BFRP structural lessons applied to Stealth-Lift chassis architecture; NSK 2025 ceramic bearing catalog data (Si ₃ N ₄ and ZrO ₂ grades); new piezoelectric linear actuator data (200 mm stroke, 50 kN force); BFRP automotive structural test data 2025; GIC threat load quantification for conventional electric forklift motor windings; battery isolation ceramic enclosure revised to ZTA (Zirconia-Toughened Alumina)

1. Executive Summary

Industrial facilities that depend on electric forklifts during post-Carrington recovery operations face a specific vulnerability that is both structural and electromagnetic. A conventional electric counterbalance forklift is built around three primary GIC-coupling elements: (1) a steel chassis and mast forming a large continuous conductive loop; (2) a copper-wound induction drive motor acting as both a GIC-induced current sink and a magnetically saturating element; and (3) a lead-acid or lithium-ion battery pack with metallic current-collection architecture exposed to the chassis ground potential.

During a Carrington-class event, the combination of these three elements means the forklift becomes an electromagnetic absorber — ground-induced currents can saturate the motor’s ferromagnetic core, destroy insulation through thermal runaway in the windings, and create arcing through the battery management system. The vehicle is rendered non-operational at precisely the moment its capabilities are most needed: moving emergency supplies, clearing debris, and supporting infrastructure recovery.

The Stealth-Lift redesign eliminates all three coupling mechanisms. Version 2.0 integrates 2025 BFRP structural performance data, updated NSK ceramic bearing specifications, and

high-force piezoelectric linear actuator capability demonstrated in 2024–2025 industrial development programs.

2. Threat Environment: GIC Vulnerability of Conventional Electric Forklifts

2.1 The Electric Forklift GIC Profile

A standard 5-ton electric counterbalance forklift presents these electromagnetic coupling pathways:

Steel Chassis Loop Area: A typical forklift chassis, approximately 2.8 m long × 1.2 m wide, forms a ground plane of ~3.4 m². During a GIC event with dB/dt = 3,000 nT/min, the induced EMF in this chassis loop is:

$$\text{EMF}_{\text{chassis}} = A \times (\text{dB}/\text{dt}) = 3.4 \text{ m}^2 \times (3,000 \times 10^{-9} \text{ T}/60 \text{ s}) = 0.17 \text{ mV}$$

Small in isolation — but the chassis is part of a larger conductive network through its steel mast (5.5 m extended height, ~6.6 m²), battery contact rails, and any metallic floor contact through the steel wheels. The effective loop area scales to 20–40 m², generating EMF of 1–2 mV. Through the low-resistance steel structure (< 0.01 Ω), this drives 0.1–0.2 A of GIC through the chassis with negligible external forcing — but this current flows directly through the battery management system’s ground reference, corrupting BMS fault detection and potentially triggering false overcurrent shutdowns.

Motor Winding Induction: The 30–50 kW AC induction motors in modern electric forklifts have copper windings in slotted silicon steel stators — a geometry specifically designed to maximize magnetic coupling. Under GIC forcing, these windings experience additional induced voltages in the 1–5 V range across the winding end turns, which are within the differential mode interference rejection capability of modern motor controllers — but the ferromagnetic stator core operates near magnetic saturation at rated load. The GIC-superimposed DC flux bias can push the core into saturation during peak load, causing:

- Asymmetric magnetization (DC offset in B-H loop)
- Harmonic distortion of the drive current (3rd harmonic up to 15% amplitude)
- Core loss increase of 20–35%
- Winding temperature rise above rated values

Net result: A conventional electric forklift may remain operational but with degraded efficiency, elevated thermal load, and BMS reliability compromise during a sustained GIC event. If the event coincides with a grid failure (the most likely scenario — GIC events destroy grid transformers), the forklift battery is depleted with no recharge available.

2.2 The Stealth-Lift Solution

By eliminating ferromagnetic structural materials, copper motor windings, and steel wheel-ground contact, the Stealth-Lift presents zero electromagnetic signature to the GIC environment. It is, electromagnetically, invisible.

3. BFRP Composite Chassis — Structure Without Steel

3.1 Basalt Fiber Reinforced Polymer (BFRP) Selection

BFRP is selected over CFRP (Carbon Fiber Reinforced Polymer) for the Stealth-Lift chassis for a specific electromagnetic reason: carbon fiber is electrically conductive ($\sigma \approx 10^3\text{--}10^4$ S/m), sufficient to support eddy current formation and GIC coupling. Basalt fiber is inherently non-conductive ($\sigma < 10^{-12}$ S/m) with mechanical properties approaching E-glass, but at higher temperature resistance and better chemical durability.

BFRP vs. CFRP vs. Steel (2025 Automotive Structural Data):

Property	BFRP (UD laminate)	CFRP (UD laminate)	Structural Steel (S355)
Density (g/cm ³)	2.1	1.55	7.85
Tensile Strength (MPa)	1,050	1,500	510
Tensile Modulus (GPa)	48	135	210
Specific Strength (MPa·cm ³ /g)	500	968	65
Electrical Resistivity (Ω·m)	$> 10^{12}$	$\sim 10^{-4}$ (conductive)	2.0×10^{-7}
Magnetic Permeability (μ_r)	1.0	1.0	~ 200
Cost (relative)	1×	4–8×	0.1×

3.2 BMW i3 Structural Lessons Applied to Stealth-Lift (V2.0 Update)

The BMW i3 electric vehicle, produced 2013–2022, employed a life module constructed from CFRP in a “carbon cage” architecture. While the i3 used CFRP for weight reduction rather than electromagnetic immunity, the structural design principles — complex load-path composite construction, bonded joint philosophy, crash energy management through composite progressive failure — are directly transferable to the Stealth-Lift BFRP chassis.

Key BMW i3 / Stealth-Lift design principle transfers:

BMW i3 CFRP Lesson	Stealth-Lift BFRP Application
Multi-layer biaxial CFRP for torsional stiffness (1,400 Nm/° measured)	BFRP $\pm 45^\circ$ bias layers for chassis torsional stiffness target: 800 Nm/°
Wet lay-up + RTM for complex node geometries	VARTM (Vacuum-Assisted RTM) for mast-to-chassis junction nodes
Energy-absorbing crush tubes (CFRP, progressive fragmentation)	BFRP crush zones at chassis ends (energy absorption: 80 kJ per zone)
Adhesive bonding primary joint method (no bolted through-thickness)	Structural epoxy bonding (Araldite® 2031 ceramic-filled) at all primary joints
40% mass reduction vs. steel equivalent	30% mass reduction achieved in BFRP Stealth-Lift equivalent

The 30% mass reduction at equivalent structural capacity translates directly to reduced battery energy requirement (lighter vehicle = less energy per operating cycle), extended oper-

ational range per charge, and reduced ground loading in post-storm environments where flooring integrity may be compromised.

3.3 Chassis Laminate Specification

BFRP Chassis Laminate Stack (primary structural sections):

[Outer Surface]

- Basalt UD 0° (300 g/m²) – primary axial load
- Basalt ±45° biaxial (300 g/m²) – shear and torsion
- Basalt UD 90° (150 g/m²) – transverse load
- Core: ZrO₂ ceramic foam (20 mm, $\rho = 0.3 \text{ g/cm}^3$) – thermal/dielectric break
- Basalt UD 90° (150 g/m²)
- Basalt ±45° biaxial (300 g/m²)
- Basalt UD 0° (300 g/m²)

[Inner Surface]

Total laminate thickness (without core): 6.5 mm

With ceramic foam core: 33 mm

Flexural rigidity (EI): 12.4 MN·m² (equivalent to 8 mm structural steel section)

Epoxy resin system: Hexion EPON™ 862 / Epikure™ W (aromatic amine cure); T_g = 175°C; no conductive filler.

4. ZrO₂ Ceramic Wheel Hubs and NSK Ceramic Bearings

4.1 Wheel Hub Design

The Stealth-Lift wheel hub must transfer all driving, steering, and braking forces from the tire/wheel to the BFRP chassis suspension arm without introducing a conductive pathway. The ground contact is through non-conductive solid polyurethane industrial tires (no steel wire reinforcement — dielectric grade specified).

ZrO₂ Wheel Hub Specification (3Y-TZP):

Parameter	Value
Material	Yttria-stabilized zirconia (3Y-TZP)
Hub bore diameter	80 mm
Flange OD	260 mm
Wall thickness (hub tube)	15 mm
Compressive strength	2,100 MPa
Flexural strength	900–1,100 MPa
Fracture toughness	8–9 MPa·m ^{1/2}
Electrical resistivity	> 10 ¹² Ω·m
Density	6.05 g/cm ³
Mass per hub	2.4 kg

4.2 NSK 2025 Ceramic Bearing Data (V2.0 Update)

NSK's 2025 Ceramic Bearing catalog documents Si₃N₄ (Silicon Nitride) and ZrO₂ grades for industrial applications requiring electrical isolation and non-magnetic operation:

NSK Ceramic Bearing Comparison — 6216 Series (80 mm bore):

Parameter	NSK 6216 Steel	NSK 6216 ZrO ₂	NSK 6216 Si ₃ N ₄
Dynamic load capacity (C)	58.5 kN	46.8 kN	51.2 kN
Static load capacity (C ₀)	37.5 kN	30.0 kN	33.5 kN
Maximum speed	5,600 RPM	4,800 RPM	7,200 RPM
Electrical resistivity	Conductive	> 10 ¹² Ω	> 10 ¹² Ω
Magnetic permeability	~200	1.0	1.0
Mass	0.72 kg	0.55 kg (□24%)	0.41 kg (□43%)
Operating temperature	□20°C to +120°C	□40°C to +200°C	□40°C to +350°C
Lubrication requirement	Standard grease	Ceramic-compatible grease	Low to none (self-lubricating)

The Stealth-Lift specifies Si₃N₄ bearings (NSK 6216 Si₃N₄ grade) at all powered wheel positions for their combination of higher dynamic load rating (relative to ZrO₂), superior temperature resistance, and self-lubricating ceramic-on-ceramic contact capability that reduces maintenance requirements in dusty industrial environments.

Bearing preload calculation for 5-tonne forklift wheel station:

Maximum dynamic wheel load (static + dynamic, loaded) = 28 kN per front wheel.

NSK 6216 Si₃N₄ dynamic load capacity C = 51.2 kN.

Load ratio: P/C = 28/51.2 = 0.547

ISO 281 bearing life equation:

$$L_{10} = (C/P)^3 \times 10^6 / (60 \times n)$$

At n = 80 RPM (wheel at 15 km/h, 600 mm diameter):

$$L_{10} = (51.2/28)^3 \times 10^6 / (60 \times 80) = 3.07 \times 10^6 / 4,800 = 639 \text{ hours}$$

With Si₃N₄ ceramic bearing life adjustment factor a₁₂₃ = 2.5 (ceramic advantage):

$$L_{10m} = 639 \times 2.5 = 1,598 \text{ operating hours}$$

At 8 hours/day operation: L_{10m} □ 200 days between preventive bearing inspections — comparable to conventional bearing service intervals.

5. Piezoelectric and Pneumatic Linear Actuators

5.1 Mast Lift Actuator — Piezoelectric Linear Design (V2.0 Update)

The V2.0 specification benefits from 2024–2025 commercial advances in high-force piezo-electric linear actuators. The Stealth-Lift mast raising and tilting system uses a modular stack of piezoelectric actuator units operating in a coordination mode:

Piezoelectric Linear Actuator Specification (2025 Commercial Grade):

Parameter	V1.0 Design	V2.0 (Commercial 2025 Unit)
Stroke (per unit)	50 mm	200 mm
Maximum force	20 kN	50 kN
Actuation speed (full stroke)	15 mm/s	40 mm/s
Power requirement	24V DC control signal	48V DC drive
Material (stack)	PZT (lead-based)	KNbO ₃ -based (lead-free, 2025)
Electrical isolation	10 kV withstand (PEEK housing)	10 kV withstand (PEEK housing)
Metal content	Minimal (titanium end caps only)	Ceramic-only active stack
Frequency response	100 Hz	200 Hz

Mast Lift System — Actuator Array Configuration:

To achieve the 5-tonne rated lift capacity, three 50 kN actuators operate in parallel on the primary mast lift axis. The parallel arrangement provides: - Total lift force: 150 kN (30.6 tonne-force) — 6.1× safety factor on 5-tonne payload - Net lift force after mast and carriage weight (1,200 kg): 138 kN — provides 5.1× safety factor - Lift speed at full load: 0.12 m/s (within industry standard 0.1–0.3 m/s range) - Full lift height (4,500 mm mast): 37.5 seconds at rated speed

5.2 Drive System — Pneumatic Bellows Actuators

The Stealth-Lift drive propulsion uses compressed nitrogen pneumatic bellows actuators in an opposed-pair ratchet drive arrangement, providing unidirectional wheel rotation from bidirectional pneumatic pressure. This eliminates all electric drive motors and their associated copper windings entirely.

Pneumatic Drive Architecture:

Each driven wheel has a pneumatic actuator assembly consisting of: - 2× PEEK-bodied pneumatic bellows cylinders (PTFE-lined, 80 mm bore, 150 mm stroke) - ZrO₂ ceramic ratchet mechanism (one-way clutch) converting linear motion to rotary - PEEK gear reduction stage (6:1 reduction from actuator to wheel) - Nitrogen reservoir (UHMWPE composite cylinder, 10 L, 200 bar)

Drive System Performance:

Parameter	Value
Maximum traction force per wheel	8.5 kN
Top speed (unloaded)	14 km/h
Top speed (5-tonne load)	9 km/h
Nitrogen reservoir endurance (full load, 8h)	180 operating minutes (requires refill)
Metal content in drive assembly	Zero (PEEK, ZrO ₂ , UHMWPE)

6. Battery Pack — ZTA Ceramic Enclosure

6.1 Zirconia-Toughened Alumina (ZTA) Enclosure (V2.0 Update)

The V1.0 battery enclosure specification used monolithic alumina (Al₂O₃). Version 2.0 upgrades to ZTA (Zirconia-Toughened Alumina) — an Al₂O₃ matrix with 15–20 vol% ZrO₂ second-phase particles that undergo stress-induced martensitic transformation at crack tips, providing substantially improved fracture toughness:

ZTA vs. Al₂O₃ for Battery Enclosure:

Property	Al ₂ O ₃ (99.9%)	ZTA (Al ₂ O ₃ + 20% ZrO ₂)
Flexural Strength (MPa)	310–380	650–750
Fracture Toughness (MPa·m ^{1/2})	3.0–3.5	5.5–6.5
Thermal Conductivity (W/m·K)	30	22
Electrical Resistivity (Ω·m)	> 10 ¹²	> 10 ¹²
Thermal shock resistance (ΔT)	180°C	280°C
Density (g/cm ³)	3.98	4.80

The improved fracture toughness of ZTA is critical for battery enclosure integrity in a fork-lift application: forklift chassis are subjected to continuous vibration, impact loading from dock levelers, and occasional collision events. The ZTA enclosure must survive these loads without fracture to maintain the battery's electromagnetic isolation from the chassis.

ZTA Enclosure Design: - Wall thickness: 12 mm (cylinder walls), 18 mm (base plate) - Joint design: Ceramic-to-ceramic butt joints sealed with ZrO₂ ceramic adhesive (Cotronics Resbond™ 940 equivalent) - Internal surface: SiN ceramic coating (50 μm) for chemical resistance to battery electrolyte - External surface: YInMn Blue enamel coat (fire-signaling color coding)

6.2 Battery Isolation Philosophy

The battery pack is maintained at a potential strictly isolated from the BFRP chassis. The isolation barrier consists of: - ZTA ceramic enclosure walls (> 10¹² Ω·m) - PEEK mounting interface brackets (> 10¹⁶ Ω·m) - Ceramic-lined UHMWPE power cables from battery to actuator control electronics

DC bus voltage isolation withstand: ≥ 10 kV (per IEC 60664-1, verified annually).

7. YInMn Blue Exterior Coating and GIC Visibility

7.1 Exterior Coating Specification

All external BFRP panels of the Stealth-Lift receive the standard CSM YInMn Blue exterior coating system:

- ZrO₂ primer (20 μ m, electrostatic spray, $\epsilon_r = 25$)
- YInMn Blue basecoat (150 μ m, HVLP, 24h cure)
- UV-stabilized polyurethane topcoat (80 μ m, gloss finish)

The YInMn Blue color serves both electromagnetic (NIR reflective, reducing solar thermal cycling of BFRP panels) and operational (standardized CSM Carrington-safe product identification) functions.

7.2 GIC Electromagnetic Signature

Stealth-Lift vs. Conventional Electric Forklift — GIC Coupling Comparison:

Parameter	Conventional Electric Forklift	Stealth-Lift
Chassis loop area	3.4 m ² (steel)	3.4 m ² (BFRP — no induction)
Inductive coupling coefficient	High ($\mu_r \sim 200$ steel)	None ($\mu_r = 1.0$ BFRP)
Motor winding sensitivity	High (copper windings + ferrite core)	None (no windings)
Battery GIC coupling	Via steel chassis ground	Isolated (ZTA enclosure)
GIC-induced thermal load	15–35°C above ambient at motor	None
Operational status during G5 event	Degraded to failed	Unaffected

8. Mass and Structural Performance Summary

Stealth-Lift vs. Steel-Frame Equivalent (5-tonne class):

Parameter	Steel-Frame Forklift	Stealth-Lift (BFRP)	Δ
Chassis mass	1,850 kg	1,295 kg	$\approx 30\%$
Total operating mass (unladen)	4,200 kg	2,940 kg	$\approx 30\%$

Parameter	Steel-Frame Forklift	Stealth-Lift (BFRP)	Δ
GIC electromagnetic coupling	High	None	—
Service life (structural)	15,000 h	20,000 h (BFRP fatigue)	+33%
Magnetic field distortion radius	2.5 m	< 0.1 m	□96%

9. Fabrication Process Flow

BFRP Prepreg Preparation (basalt UD, biaxial, sizing verification)
 □
 Chassis Mold Lay-Up (VARTM, 120°C cure, 8h)
 □
 ZrO□ Ceramic Foam Core Bonding (Araldite 2031, 24h cure)
 □
 Secondary Laminate Lay-Up (inner faces)
 □
 CNC Trimming and Drilling (diamond tooling only)
 □
 ZrO□ Hub Fabrication (CIP □ Sinter 1450°C □ Diamond Grind)
 □
 Si□N□ Bearing Installation (PEEK housing, ceramic-grade grease)
 □
 Pneumatic Actuator Assembly (PEEK cylinders, ZrO□ ratchets)
 □
 Piezoelectric Mast Actuator Array Integration
 □
 ZTA Battery Enclosure Manufacture (CIP □ Sinter □ Joint Sealing)
 □
 Battery Pack Installation and Isolation Verification (10 kV withstand)
 □
 BFRP Panel Assembly (structural adhesive bonding, jig-controlled)
 □
 YInMn Blue Coating System Application
 □
 Load Test (5-tonne capacity, 1.5× proof load) + GIC Isolation Test

10. Quality Acceptance Criteria

Parameter	Specification	Test Method
Chassis torsional stiffness	□ 800 Nm/°	Static torsion fixture
Wheel hub compressive load	No fracture at 2× WLL	Static load test
Bearing L ₁₀ m life	□ 1,500 h	Calculated per ISO 281
Battery isolation withstand	□ 10 kV DC, 1 min	IEC 60664-1

Parameter	Specification	Test Method
Chassis electrical resistivity	$> 10^{10} \Omega$ (end-to-end)	ASTM D257
Actuator lift force	$\square 100$ kN (3-unit array)	Load cell verification
YInMn Blue NIR reflectance	$\square 38\%$ at 800 nm	Spectrophotometer
Proof load test	7.5 tonne (1.5 $\times$ rated)	Fork load test

11. References and Standards

- ISO 281: Rolling Bearings — Dynamic Load Ratings and Rating Life
- NSK Ceramic Bearing Catalog 2025, NSK Ltd.
- BMW i3 CFRP Life Module Design — Technical Documentation (BMW Group, open publication)
- ASTM D257: Standard Test Methods for DC Resistance or Conductance of Insulating Materials
- IEC 60664-1: Insulation Coordination for Equipment Within Low-Voltage Supply Systems
- BFRP Mechanical Properties Review, Basalt Fiber & Composite Materials Technology Development Co., 2024–2025
- Piezoelectric Industrial Linear Actuator Technical Specifications, 2025 commercial catalog data
- “Geomagnetic Storm Effects on Variable Frequency Drive Systems,” *IEEE Transactions on Power Electronics*, 2024

“An electromagnetic storm does not need a direct lightning strike to disable your equipment. The storm can reach through the floor, through the steel frame, through the motor windings. The Stealth-Lift has nothing for it to reach through.”

— **CSM Engineering Design Directive, Aegis Home Series**

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